

Challenge 04

The Task:

- Build an **AI planning agent (The Tactician)** that receives a Work Order and creates an optimised RFQ vendor selection strategy.
- The agent must analyse vendor health, market conditions, commodity prices, and competitor activity before making decisions.
- It should adapt the standard approach when the vendor pool is unhealthy, capacity is constrained, prices are rising, or urgency is high.
- Complete the **system prompt, goal-oriented agent loop, and structured RFQ plan extraction** in `starter.py`.
- Test the agent across three different time periods and compare whether its strategy adapts appropriately to changing conditions.

Input:

Data:

`work_order_id` - Workorder the agent needs to run for

`month_index` - Month the agent has to run for

Following tools are available for the agent in `starter.py`

- Work Order details, category, budget, and urgency
- Vendor pool for the required category
- Vendor health score and trend
- Market signals
- Commodity price history
- Competitor capacity/activity signals from Hexagon AG

Briefing.md that describes your mission to be completed

starter.py that contains the Boilerplate Code

Output:

1. Complete `starter.py` with necessary code that generates 4 graphs:
2. A structured **RFQPlan** containing:
 - Work Order ID, category, urgency, and budget ceiling
 - Prioritised vendor shortlist with health scores and rationale
 - Number of vendors to invite
 - RFQ terms, including bid window and evaluation weights

- Market context and risk flags
- Adaptation notes explaining deviations from the standard RFQ strategy

The challenge also requires generating and comparing plans for **Month 5, Month 10, and Month 15** to demonstrate adaptive decision-making.

Solution:

There are many ways to arrive at the solution, in this workshop, we can try out two options:

Joule Desktop

Claude Code

Joule Desktop

Open Joule Desktop -> New Conversation

Select SAP HANA as connector (*ensure that connector is in connected state, if not follow the set up guide.*)

Attach Briefing.md and starter.py to the conversation

Enter the following prompt

```
Attached Briefing.md contains the challenge. use HANA database to analyse the data instead of CSVs. Complete the starter.py code. Rewrite, Create and Deploy this joule desktop skill and run the same
```

Let it run and complete the task. You should see the following output:

Results

RFQ Plan — WO-2024-0078

Category: Mechanical Parts **Urgency:** Expedited **Budget Ceiling:** \$40,739.55 **Procurement Manager:** Leena Mathew **Month:** 7 (August 2024)

RFQ Terms

- **Bid window:** 6 days (*expedited urgency baseline*)
- **Evaluation weights:** Price **0.30** | Delivery **0.40** | Quality **0.20** | Relationship **0.10**
- **Special conditions:** Commodity shock adjustment applied across all 5 Mechanical Parts inputs — delivery weight elevated, price weight reduced. Acceleration risk acknowledged for expedited timeline.

Vendor Exclusions

1 vendor hard-excluded:

- **V022 Andean Precision SRL** — Health score 64.9 (< 65) AND trend 'declining' → hard exclusion rule #2. Also carrying alert flag "declining_trend". Removed before scoring.

Pool after exclusions: 8 vendors → all included (meets standard shortlist target).

Market Context

All **5 Mechanical Parts commodities** are showing active shock conditions over the last 8 weeks (July–August 2024):

Commodity	Latest Price	Shock Weeks	Max Shock Premium
Steel HR Coil	\$683.94	8/8	+8.5%
Aluminium 6061	\$2,562.65	8/8	+8.5%
Stainless 316L	\$3,547.93	8/8	+8.5%
Ball Bearing 6205	\$4.05	7/8	+5.0%
Titanium Sponge	\$14.94	7/8	+5.0%

Additionally, the **Port of Hamburg was hit by a 3-week dock worker strike** (signal: July 4, 2024; sentiment −0.8; cascade potential 0.75) — directly threatening European vendors V004, V009, V001, and V019's logistics chains. The **Cerro Negro Metals Q2 financial report** (August 15, 2024; Reuters) signals aggressive price cutting ahead from a major steel/aluminium supplier, introducing pricing instability.

Risk Flags

1. **V017 Türk Makina A.Ş. — OTD Collapse Risk:** Health score is 100.0 (improving) but live OTD rate at month 7 is only **66.7%**. This is a critical tension for an expedited order where delivery weight is 0.40. Include for bidding breadth; do NOT assign delivery-critical line items.
2. **V024 BalticTech UAB — New Vendor Execution Risk:** OTD = **0.0%** in latest performance snapshot (3 historical orders; thin data). Include for price competition only; execution reliability is unproven.

3. **V001 Kovacs Precision GmbH — Relationship Declining:** Vendor master shows relationship arc 'declining' (9 years, but downward trajectory). Strategic withdrawal or reduced commitment risk. 1 open dispute (\$7,901). Monitor for bid quality.
 4. **Broad Commodity Shock — Cost Pass-Through Risk:** All 5 input commodities are in sustained shock (7–8 consecutive shock weeks). Vendors will likely embed cost premiums of +5–8.5% in bids. Price evaluation weight reduced to 0.30 in response.
 5. **Port of Hamburg Backlog — European Logistics Risk:** Ongoing dock worker strike (July 2024) affects the logistics chains of 4 shortlisted European vendors (V004 Strömberg, V009 Fischer, V001 Kovacs, V019 Emerald Isle). Request explicit lead time confirmation from each.
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Adaptation Notes

- **Commodity shock trigger → weights adjusted:** All 5 Mechanical Parts commodities have ANY_SHOCK = 1 (sustained 7–8 shock weeks, premiums of +5–8.5%). Applied mandatory adaptation: price weight 0.40 → **0.30**; delivery weight 0.30 → **0.40**.
- **Urgency = expedited:** Bid window set to 6 days (from 7-day baseline). Acceleration risk flagged — limited time for vendor due diligence on commodity-adjusted pricing.
- **Competitor activity = NORMAL:** Month 7 is below the Hexagon AG high-activity threshold (months 11–17). No bid window extension applied for competitor pressure.
- **V007 Nakamura Tooling Co. preference override:** Despite declining health trend (91.1), included ahead of others due to exceptional quality score (9.47 — best in pool) and 100% OTD. Quality-critical items should be allocated here.

Claude Code

Open a new terminal at RFQ folder

Type *Claude*, once claude comes up, enter the following prompt:

```
Read the BRIEFING.md from challenges/challenge_04_agent_tactician folder,
use the csv data from output_v2 folder and Complete the code for starter.py.
Complete the starter.py code. Rewrite, Create and Deploy this as skill and
run the same
```

Wait for claude to complete the task

Results

Field	Value
Category	Mechanical Parts
Urgency	Critical (Nov 13 delivery deadline)
Budget ceiling	\$10,433 (+15% expedited pre-approved = \$11,997)
Vendors shortlisted	8

Vendor shortlist (ranked):

1. Adriatic Fastenings d.o.o. – score 92.4, stable
2. Emerald Isle Castings Ltd – score 91.4, stable
3. Fischer & Söhne KG – score 88.9, declining (caution)
4. Strömberg Hydraulics AB – score 87.8, declining (caution)
5. Andean Precision SRL – score 86.3, stable, new_proving
6. Türk Makina A.Ş. – score 84.9, stable
7. Nakamura Tooling Co. – score 78.7, alert ⚠ (mitigation plan required)
8. BalticTech UAB – score 77.8, alert ⚠ (capacity filler only)

Key adaptations from standard playbook:

- V001 (Kovacs) excluded despite score 73.8 – payment reliability of 54 with alert_flag is unacceptable on a critical order
- Delivery weight raised to 0.4 (from 0.3) and price reduced to 0.3 – critical urgency drives delivery priority
- Special conditions added: written mitigation plans for alert vendors, fixed delivery commitment language, expedited premium pre-approval

Risk flags: 6 named risks including a critical delivery timing warning (7-day bid window equals the delivery lead time for item I01).